$$\bar{x} = \frac{\sum_{i=1}^{8} x_i}{n} = \frac{368}{8} = 46$$

$$s^2 = \frac{\sum_{i=1}^{8}(x_i - \bar{x})^2}{(n-1)} = \frac{\sum_{i=1}^{8}(x_i - 46)^2}{(8-1)} = \frac{162}{7} = 23.143.$$

The calculated value of chi-square is $\chi_0^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{\sum_{i=1}^{8}(x_i - \bar{x})^2}{\sigma_0^2} = \frac{162}{48} = 3.375$

Step 6 : Critical values

Since H_1 is a two sided alternative, the critical values at α =0.05 are $\chi^2_{7,\,0.025} = 16.01$ and $\chi^2_{7,0.975} = 1.69$.

Step 7 : Decision

Since it is a two-tailed test, the elements of the critical region are determined by the rejection rule $\chi_0^2 \geq \chi^2_{n-1,\frac{\alpha}{2}}$ or $\chi_0^2 \leq \chi^2_{n-1,1-\frac{\alpha}{2}}$.

For the given sample information, the rejection rule does not hold, since

$1.69 = \chi^2_{7,\,0.975} < \chi_0^2\ (=3.375) < \chi^2_{7,0.025} = 16.01.$

Hence, H_0 is not rejected in favour of H_1. Thus, Population variance can be regarded as 48 kg.

Example 2.7

A normal population has mean μ (unknown) and variance 9. A sample of size 9 observations has been taken and its variance is found to be 5.4. Test the null hypothesis H_0: $\sigma^2 = 9$ against H_1: $\sigma^2 > 9$ at 5% level of significance.

Solution:

Step 1 : Null Hypothesis H_0: $\sigma^2 = 9$.

i.e., Population variance regarded as 9.

Alternative hypothesis H_1: $\sigma^2 > 9$.

i.e. Population variance is regarded as greater than 9.

Step 2 : Data

Sample size (n) = 9

Sample variance (s^2) = 5.4

Step 3 : Level of significance

α = 5%

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